Power Management Challenges in Linux*

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Intel Open Source Technology Center

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What Power Management Is About

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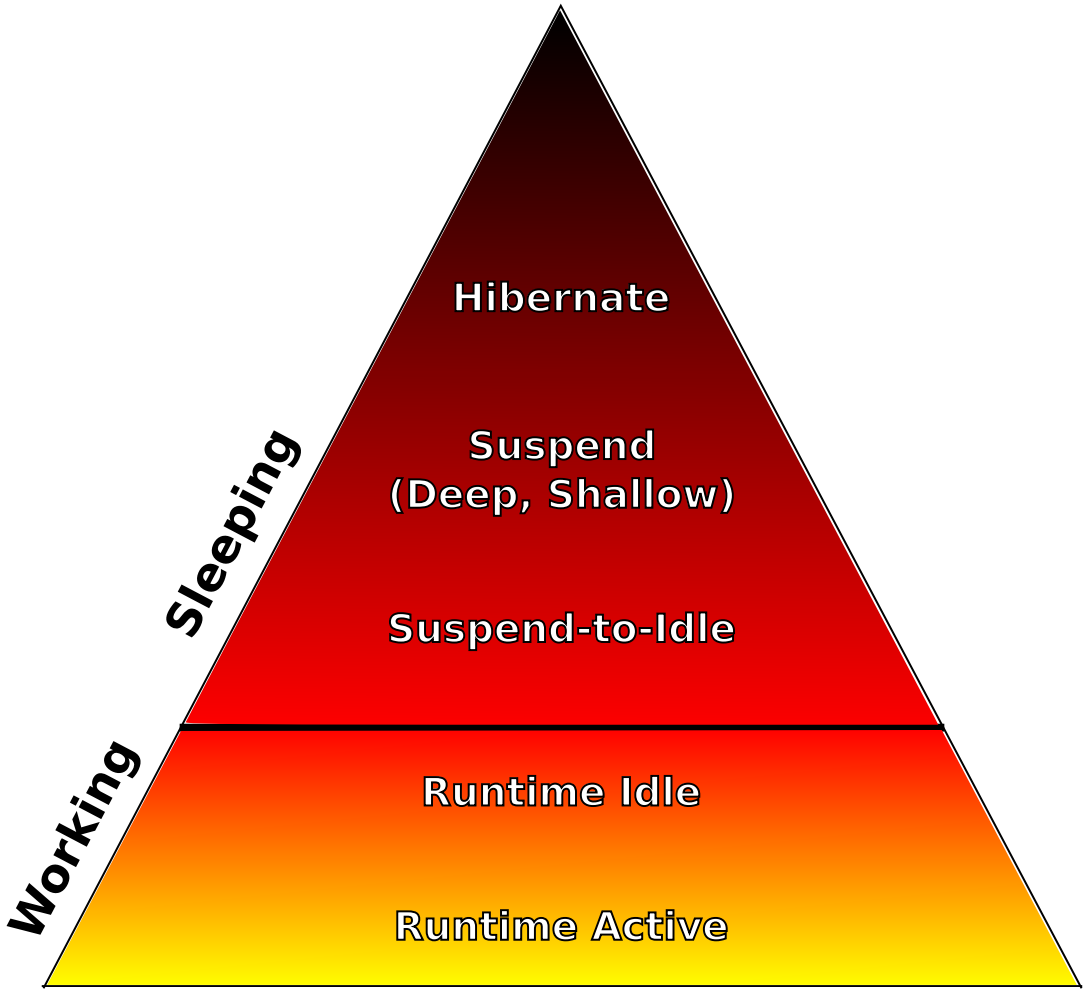
Linux Power Management Overview



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System-Wide PM (Sleep States) and Working-State PM

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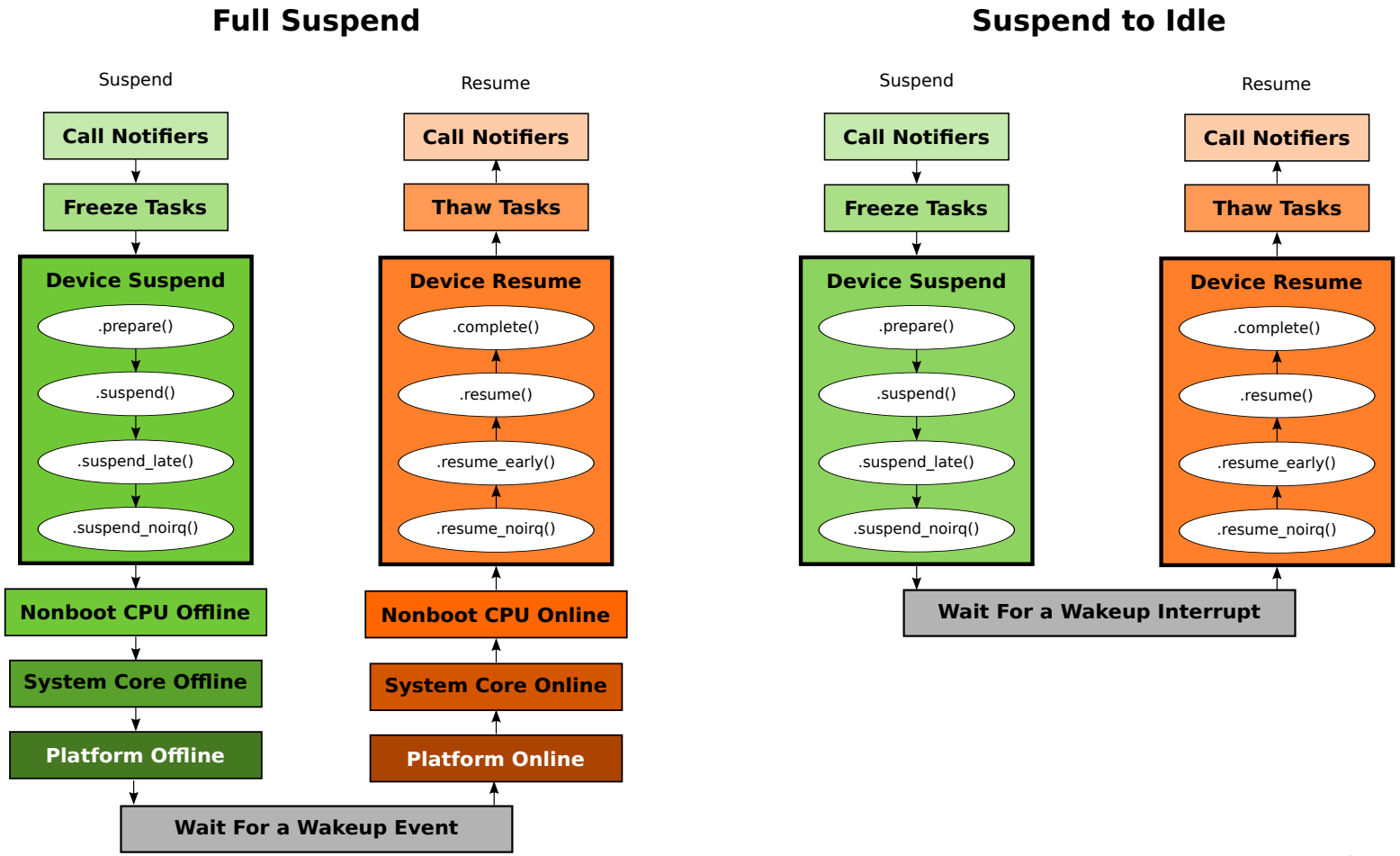
System Suspend Control Flows



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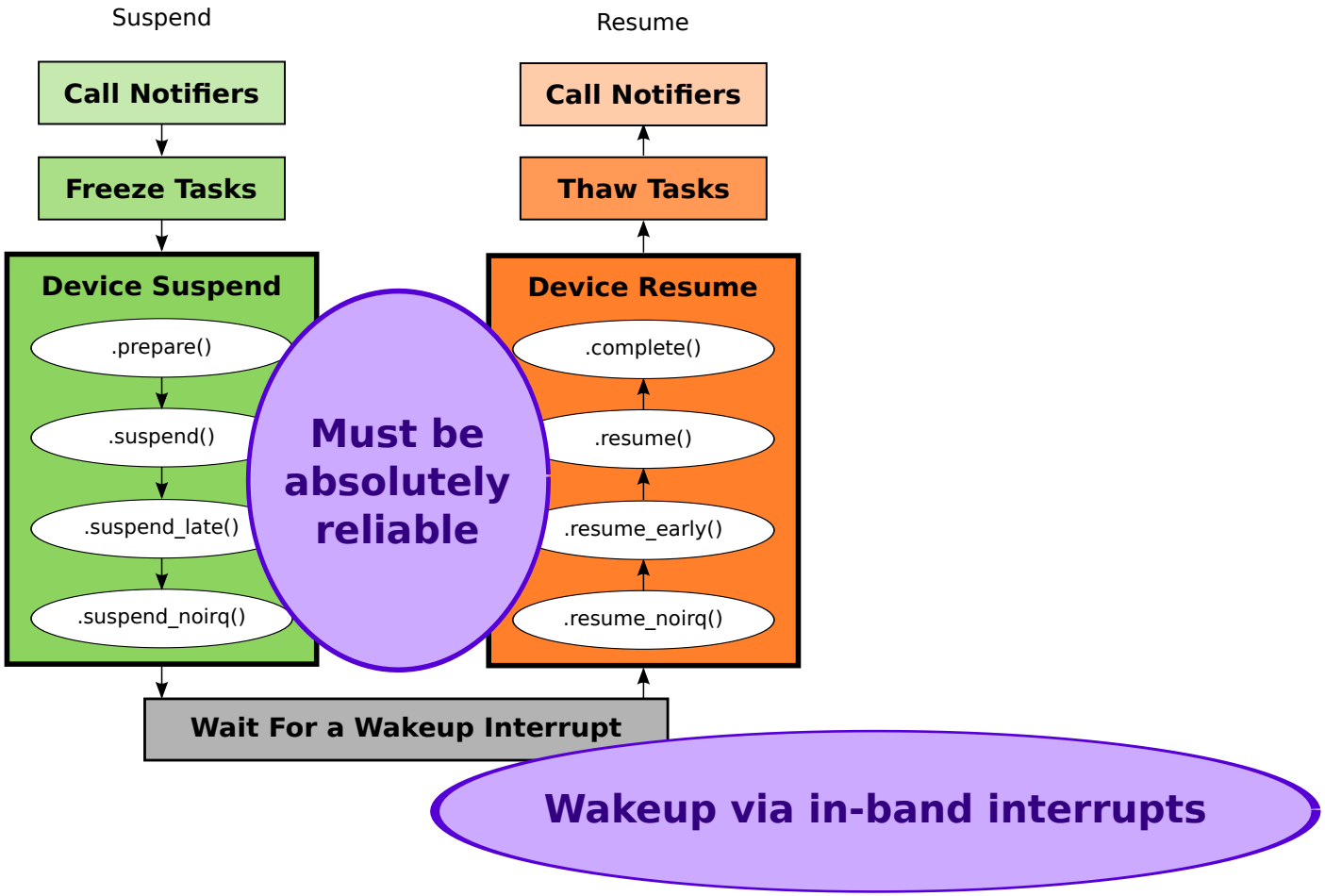
Suspend-to-Idle Implementation Challenges



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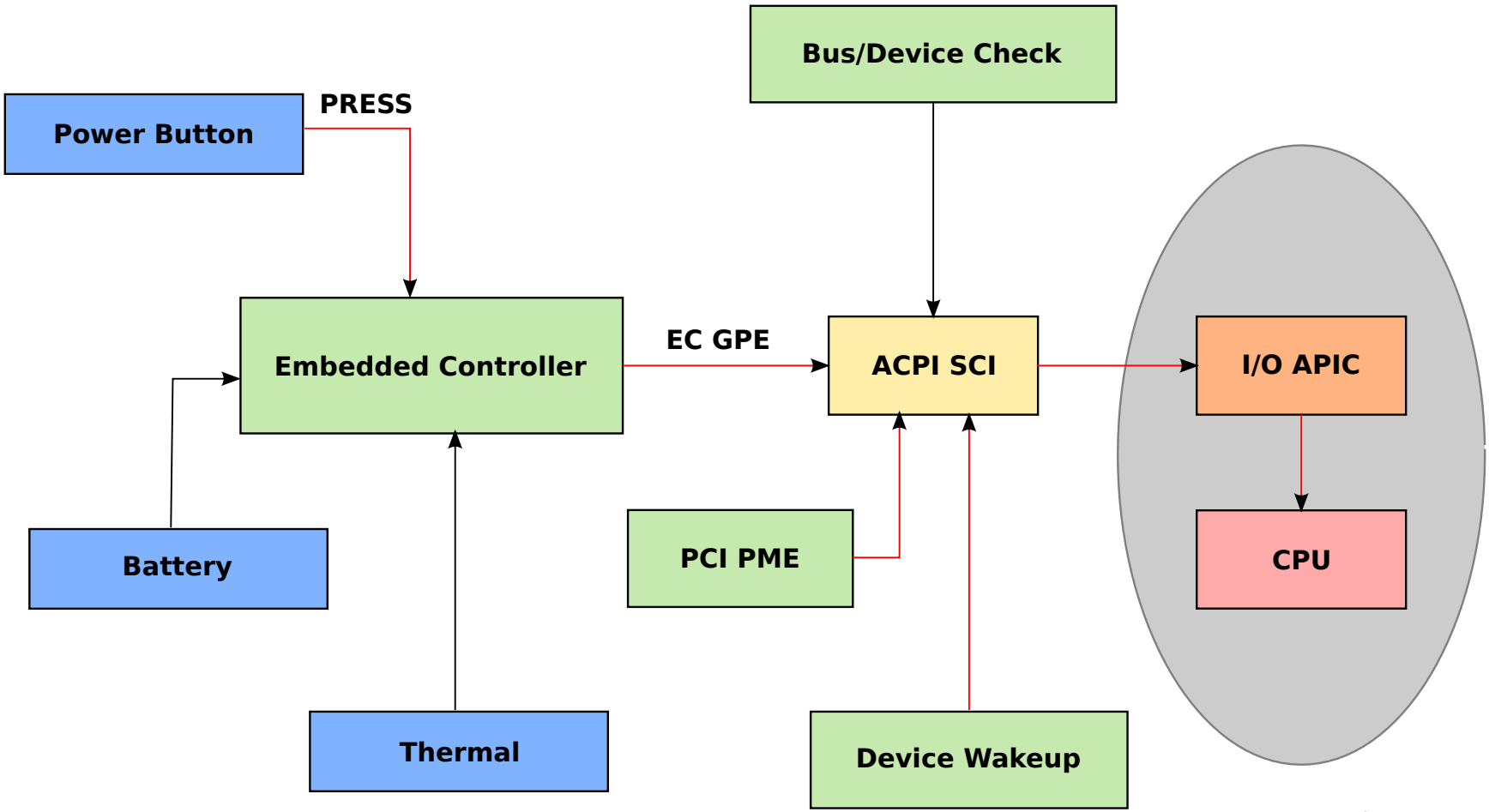
Example: EC-Based Power Button Events Signaling



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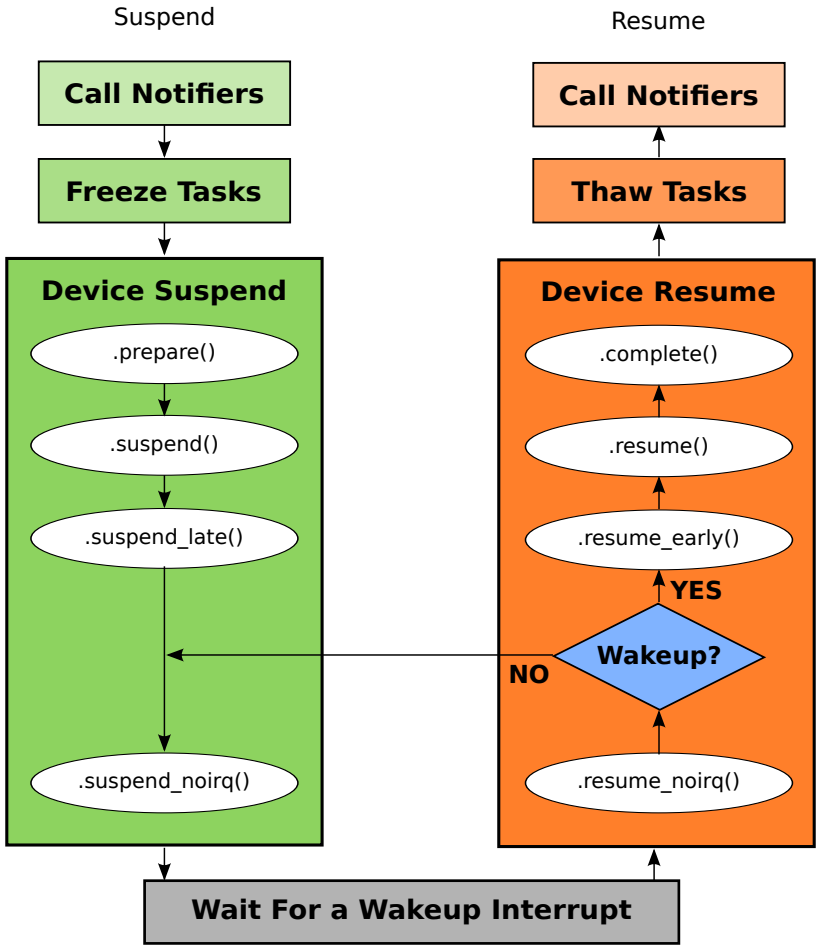
Suspend-to-Idle With Detection of Spurious Wakeups



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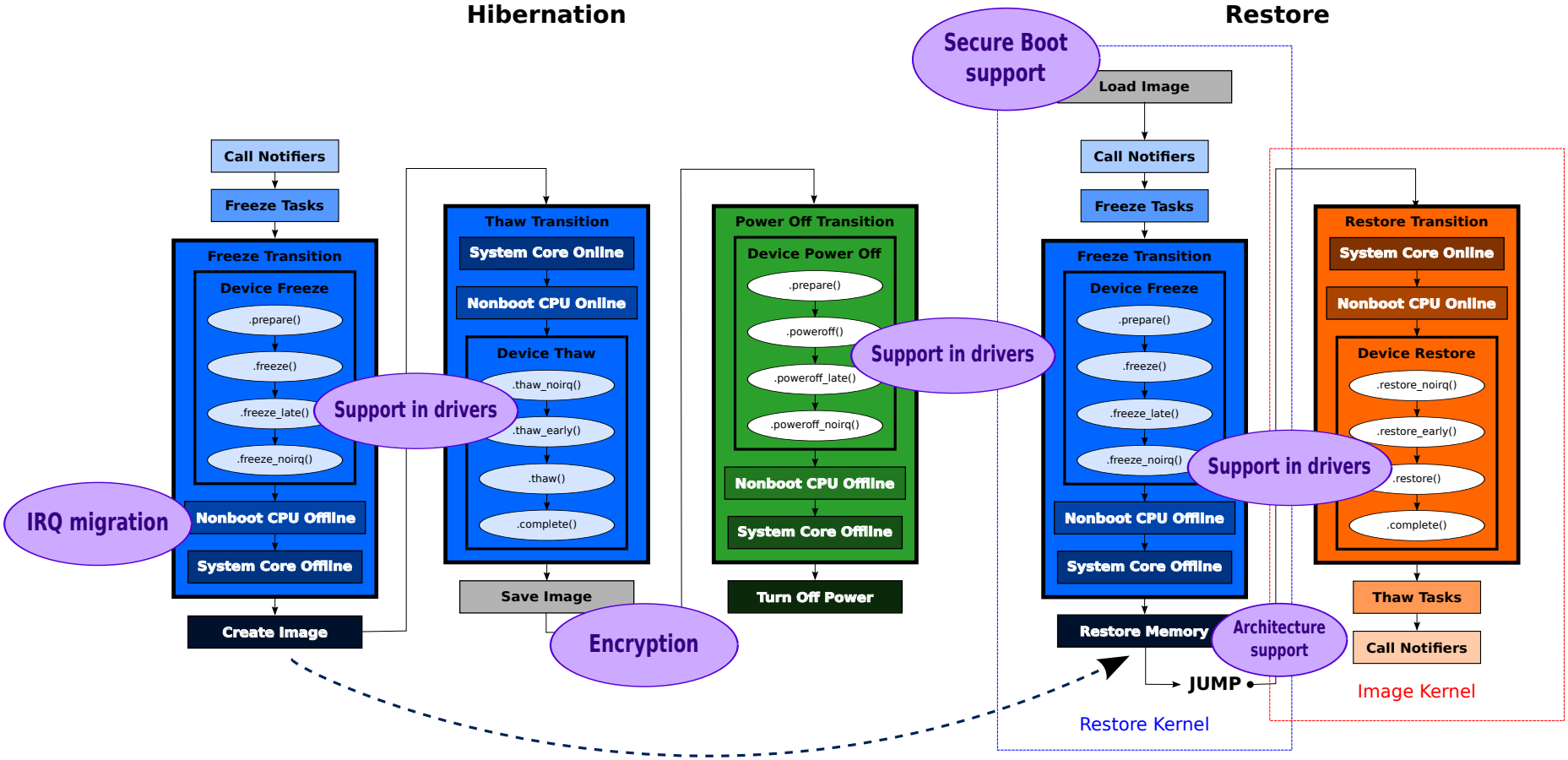
Challenges Related to Hibernation



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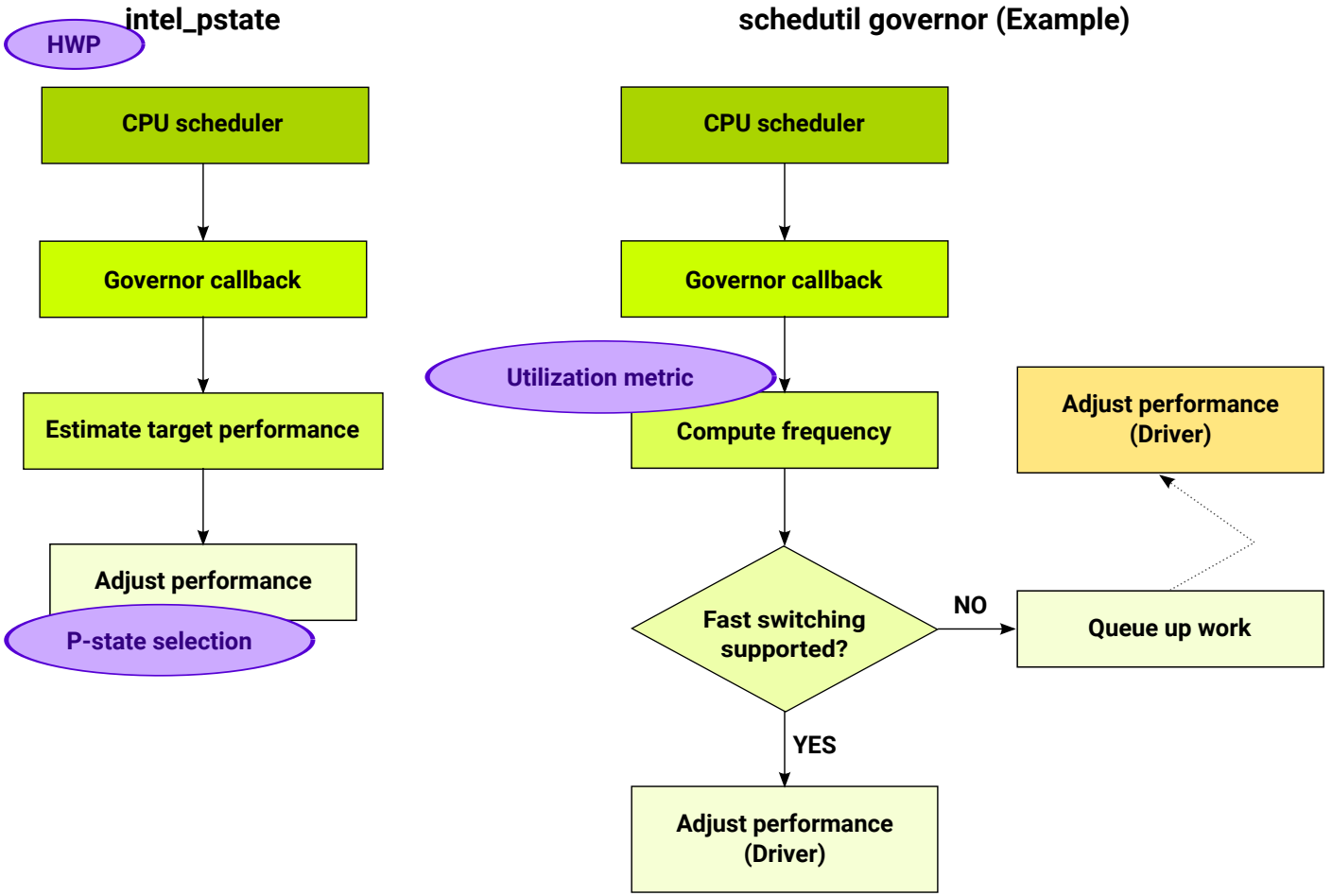
CPU Performance Scaling



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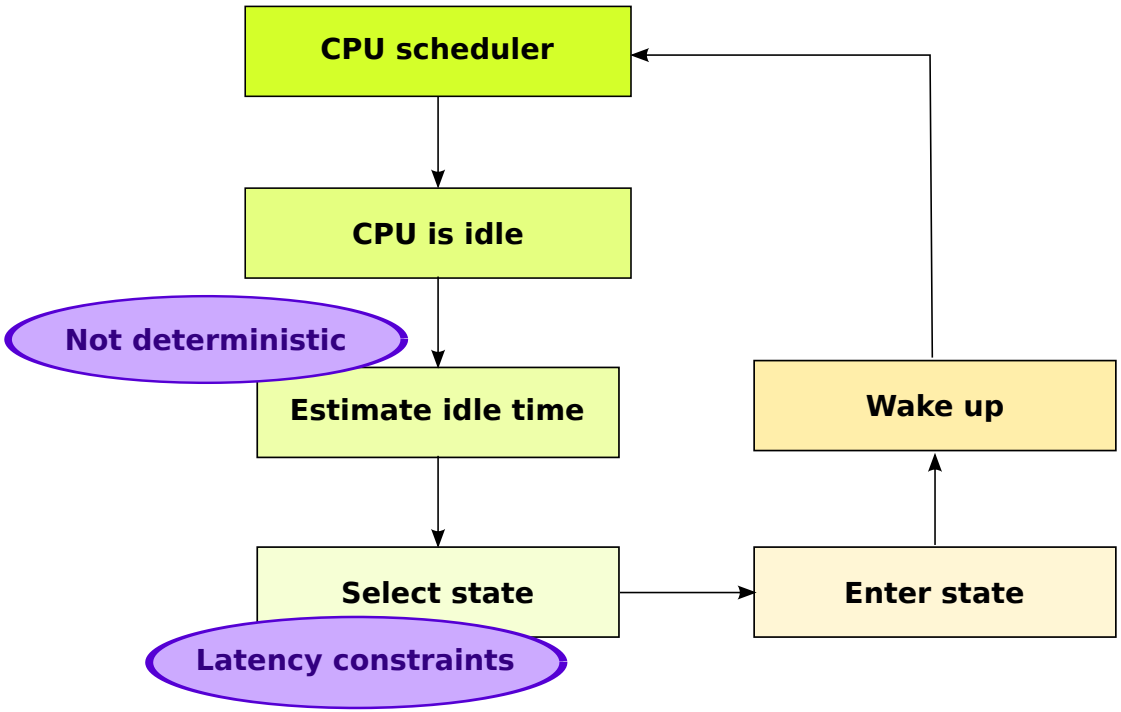
CPU Idle Time Management



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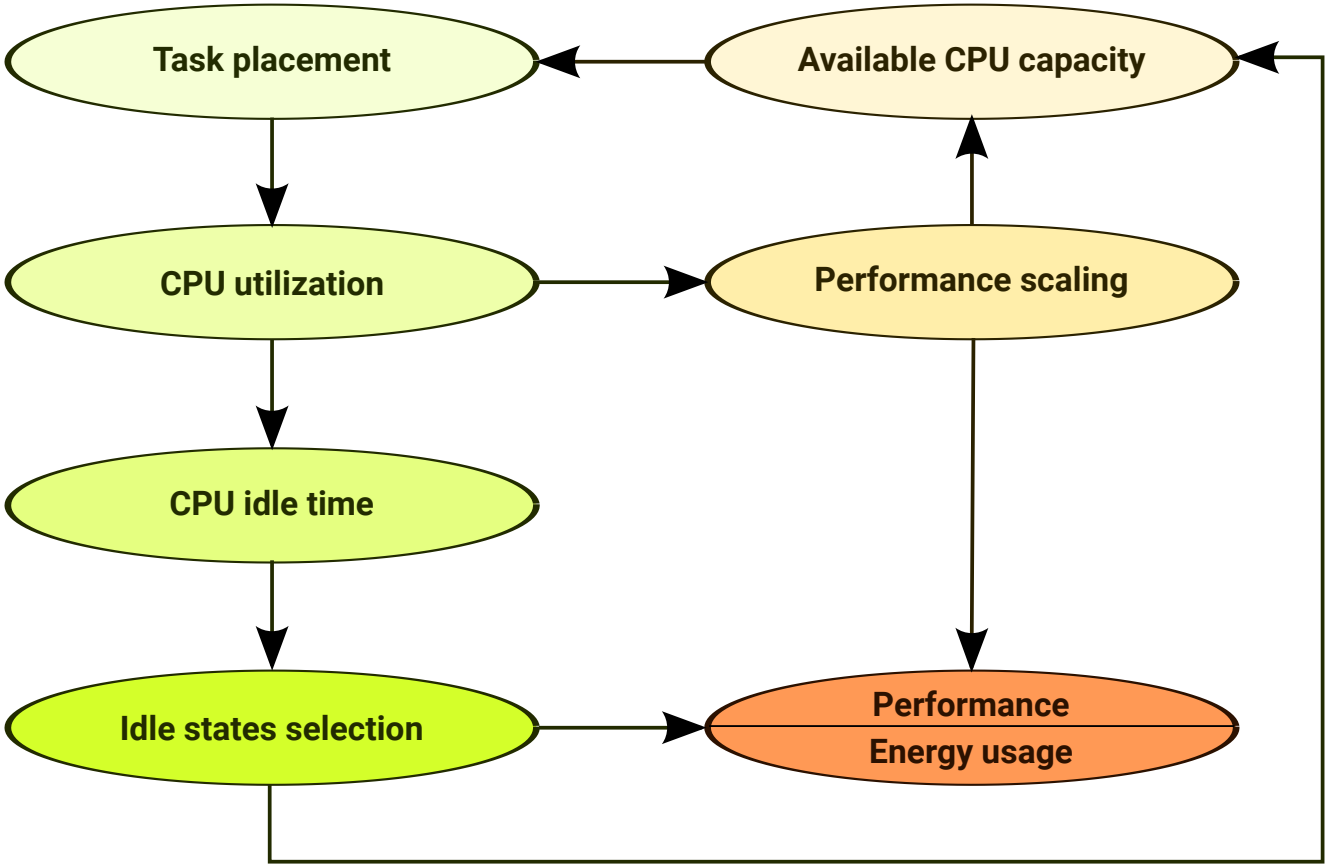
Task Placement and CPU PM



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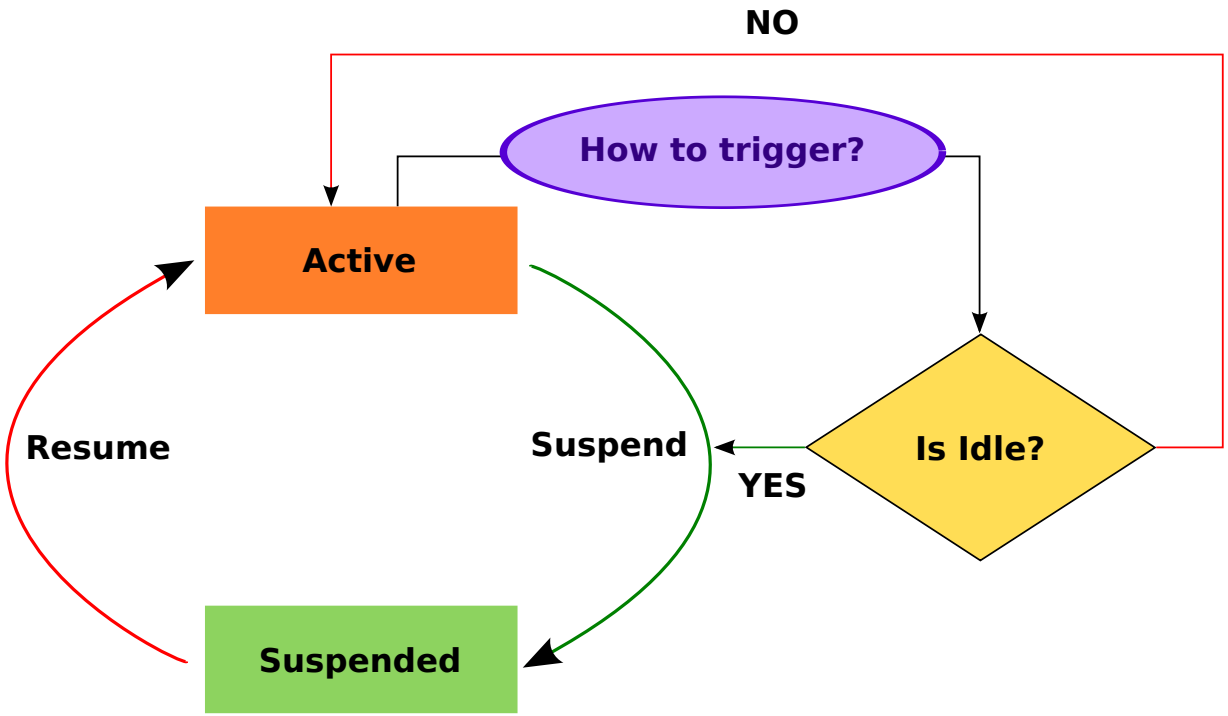
Device Runtime PM Framework



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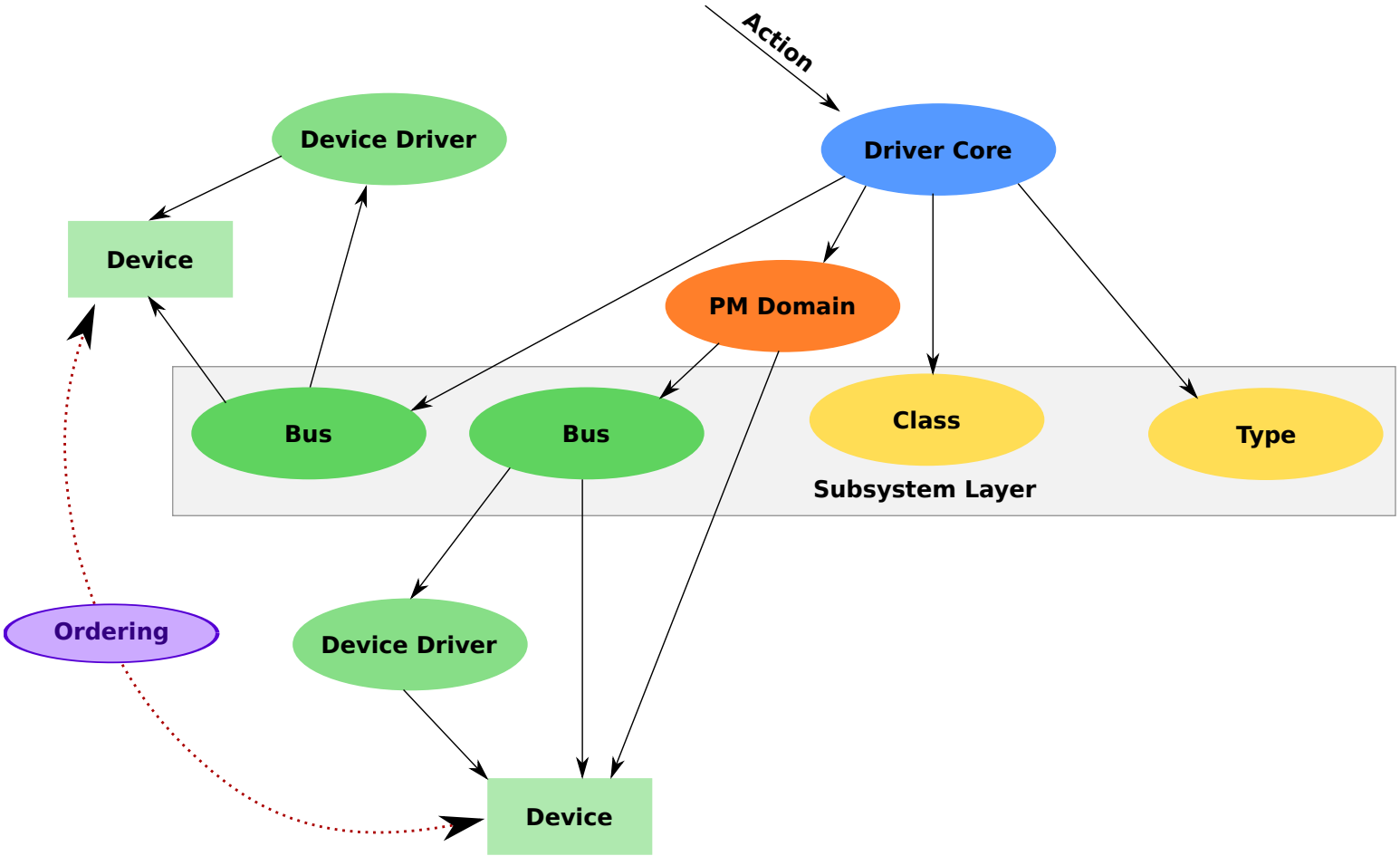
Driver Core and Device PM Operations



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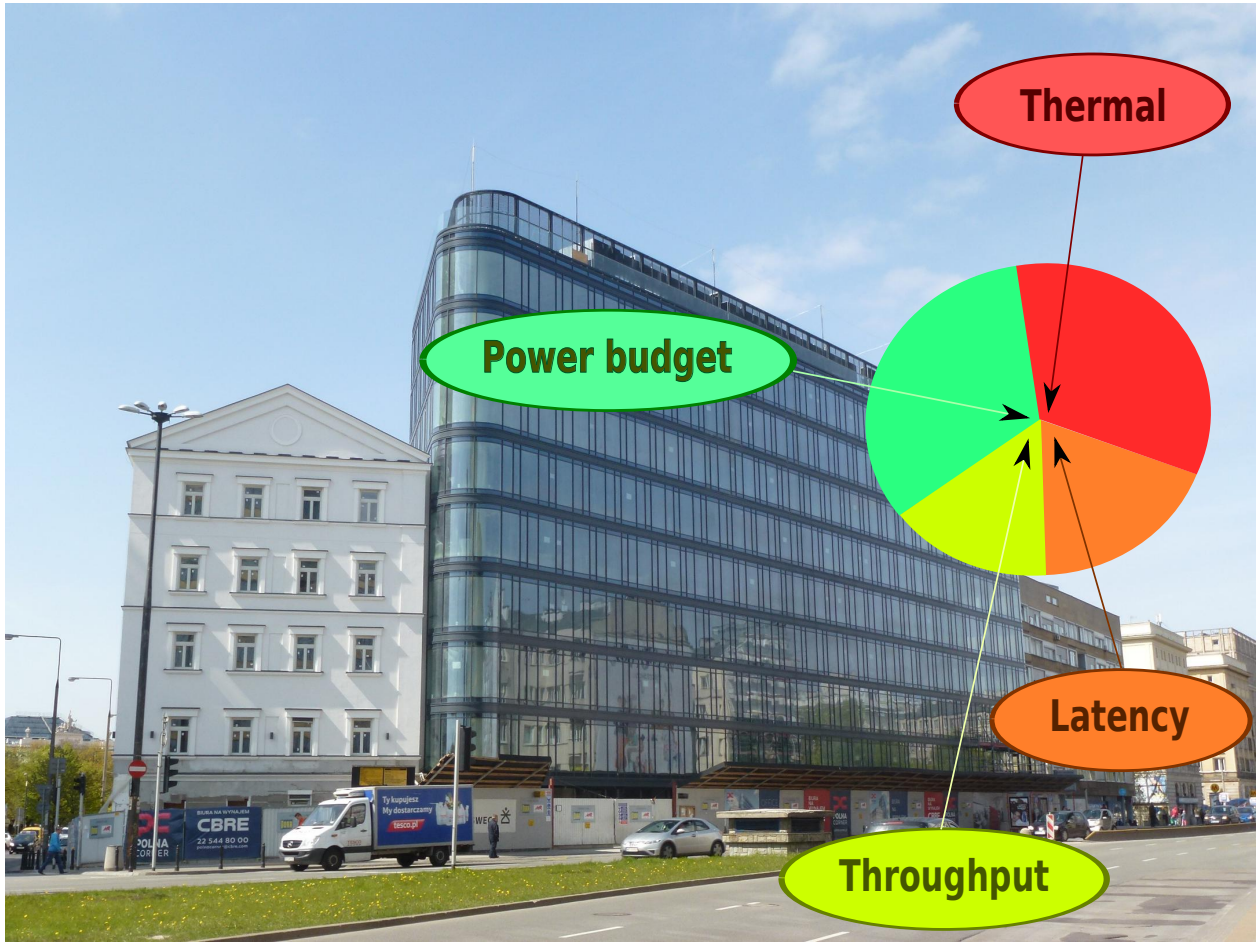
Constraints and Requirements



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Questions?

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References

R. J. Wysocki, *PM Infrastructure in the Linux Kernel – Current Status and Future* (https://events.linuxfoundation.org/sites/events/files/slides/kernel_PM_infra_0.pdf).

R. J. Wysocki, *CPUfreq and The Scheduler: Revolution in CPU Power Management* (https://events.linuxfoundation.org/sites/events/files/slides/cpufreq_and_scheduler_0.pdf).

Len Brown, *Suspend/Resume at the Speed of Light* (http://events.linuxfoundation.org/sites/events/files/slides/Brown-Linux-Suspend-at-Speed-of-Light-LC-EU-2015.pdf).

R. J. Wysocki, *What Is Suspend-to-Idle and How To Make It Work* (http://events.linuxfoundation.org/sites/events/files/slides/what-is-suspend-to-idle.pdf).

R. J. Wysocki, *Getting More Out Of System Suspend In Linux* (http://events.linuxfoundation.org/sites/events/files/slides/linux_suspend.pdf).

R. J. Wysocki, *Power Management in the Linux Kernel – Current Status and Future* (http://events.linuxfoundation.org/sites/events/files/slides/kernel_PM_plain.pdf).

R. J. Wysocki, *Why We Need More Device Power Management Callbacks* (https://events.linuxfoundation.org/images/stories/pdf/lfcs2012_wysocki.pdf).

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